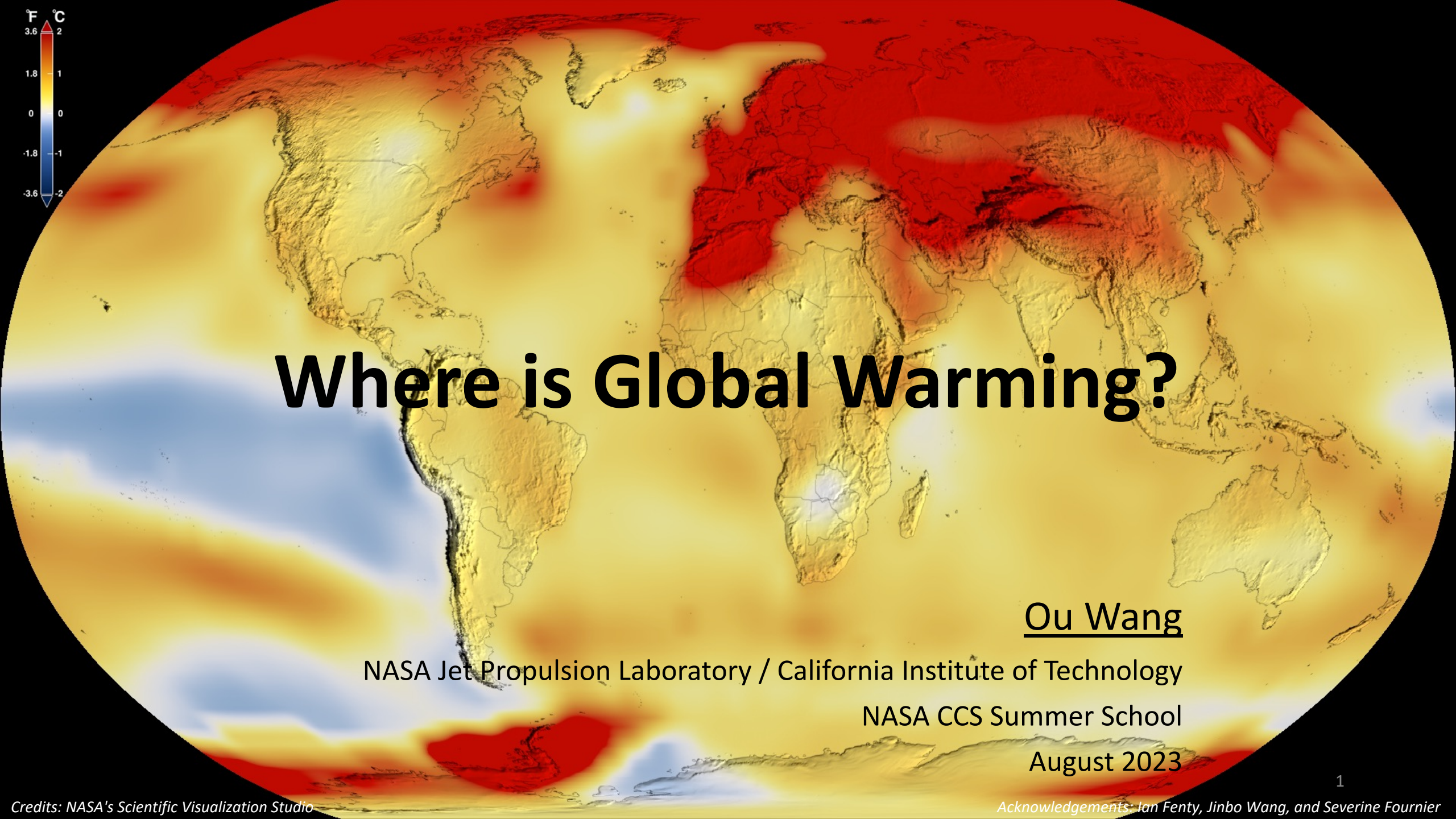


A world map showing temperature anomalies. The map uses a color scale from blue (cooler) to red (warmer). Significant red areas are visible in the Arctic region, parts of Europe, Asia, and the Southern Ocean. A color scale bar is located in the top left corner, showing values from -3.6 to 3.6 in both Fahrenheit and Celsius.

Where is Global Warming?

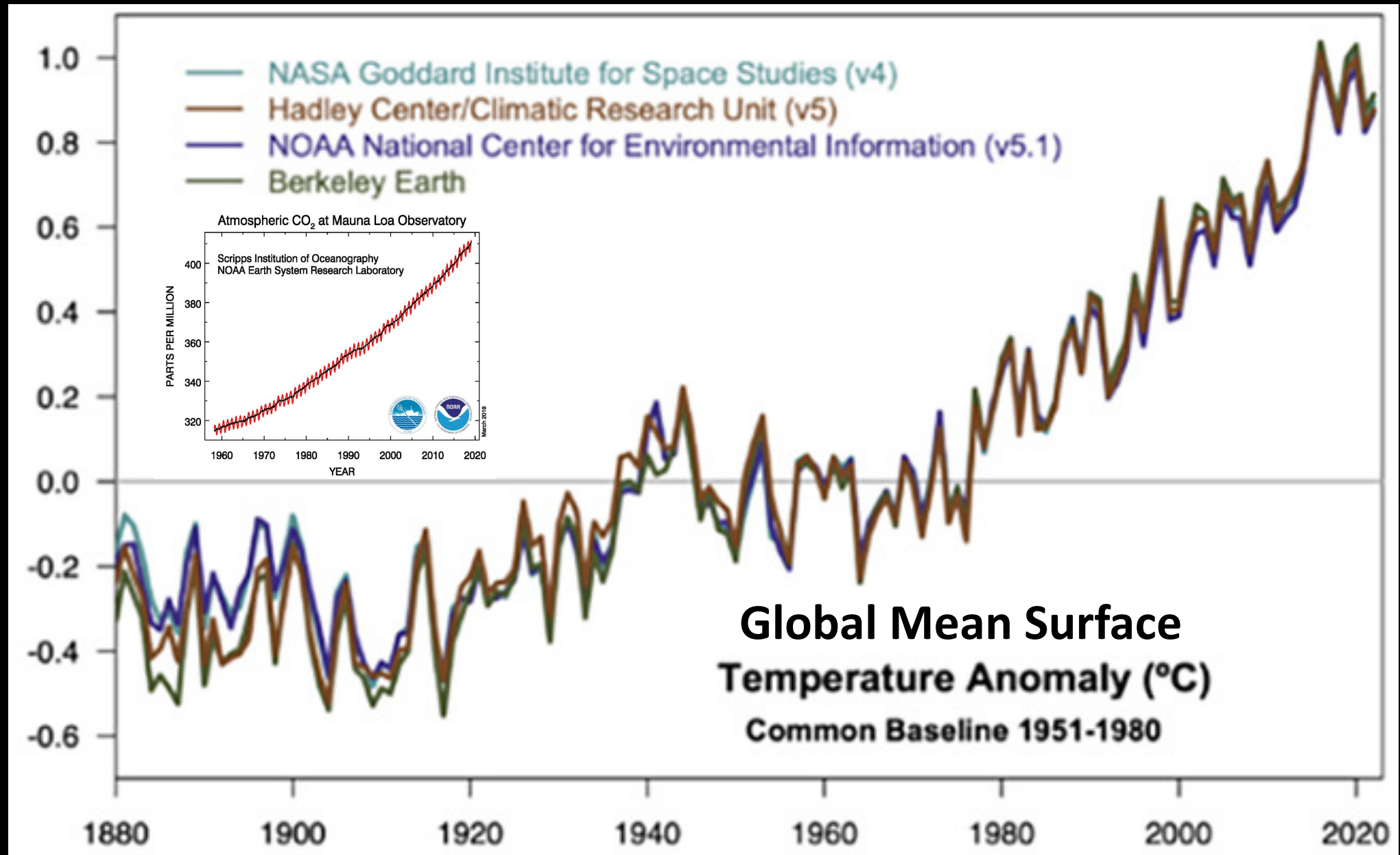
Ou Wang

NASA Jet Propulsion Laboratory / California Institute of Technology

NASA CCS Summer School

August 2023

Earth is warming



July 2023, the warmest month on record

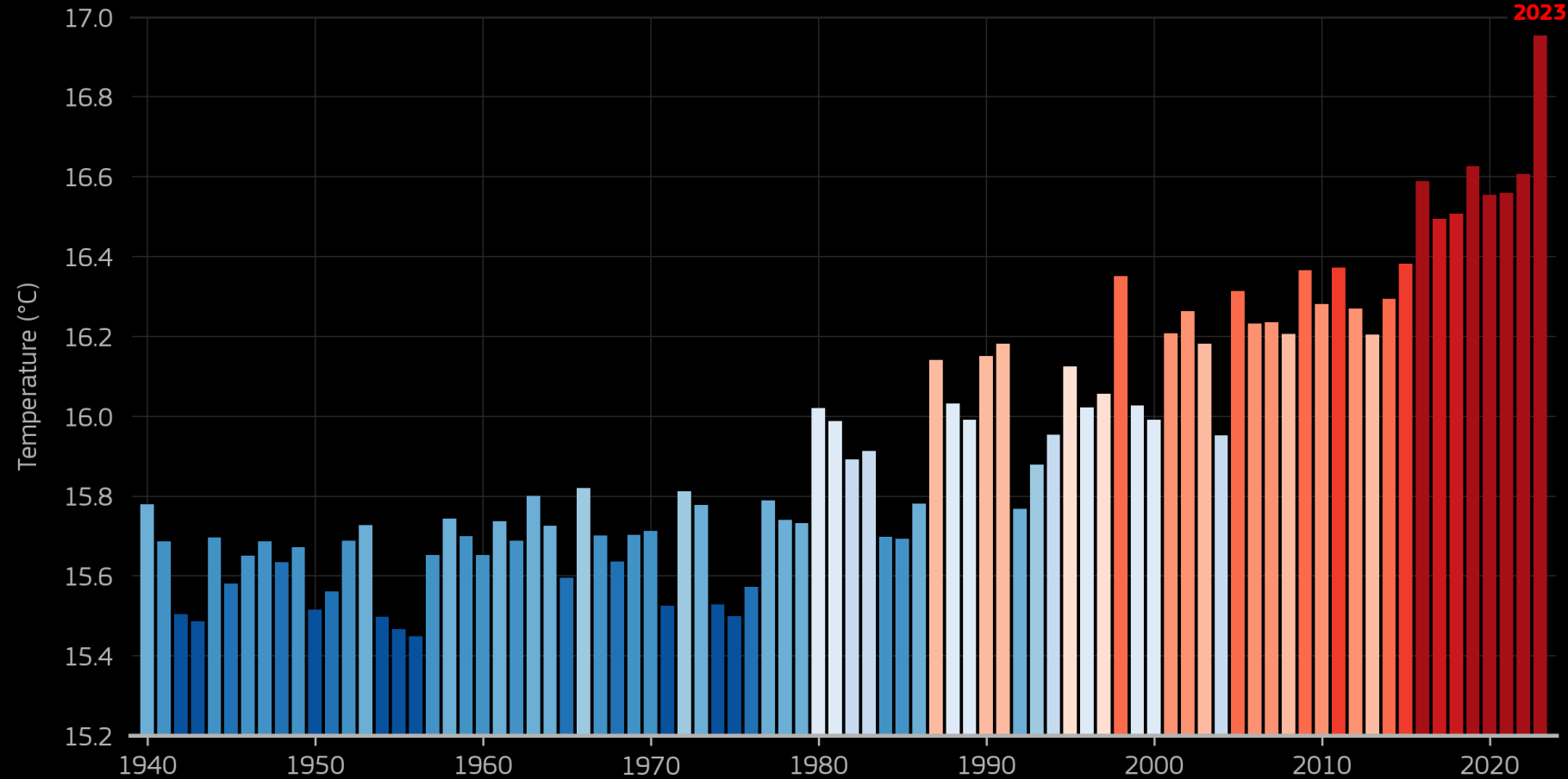
GLOBAL SURFACE AIR TEMPERATURE • JULY

Data: ERA5 1940–2023 • Credit: C3S/ECMWF



Climate
Change Service

climate.copernicus.eu



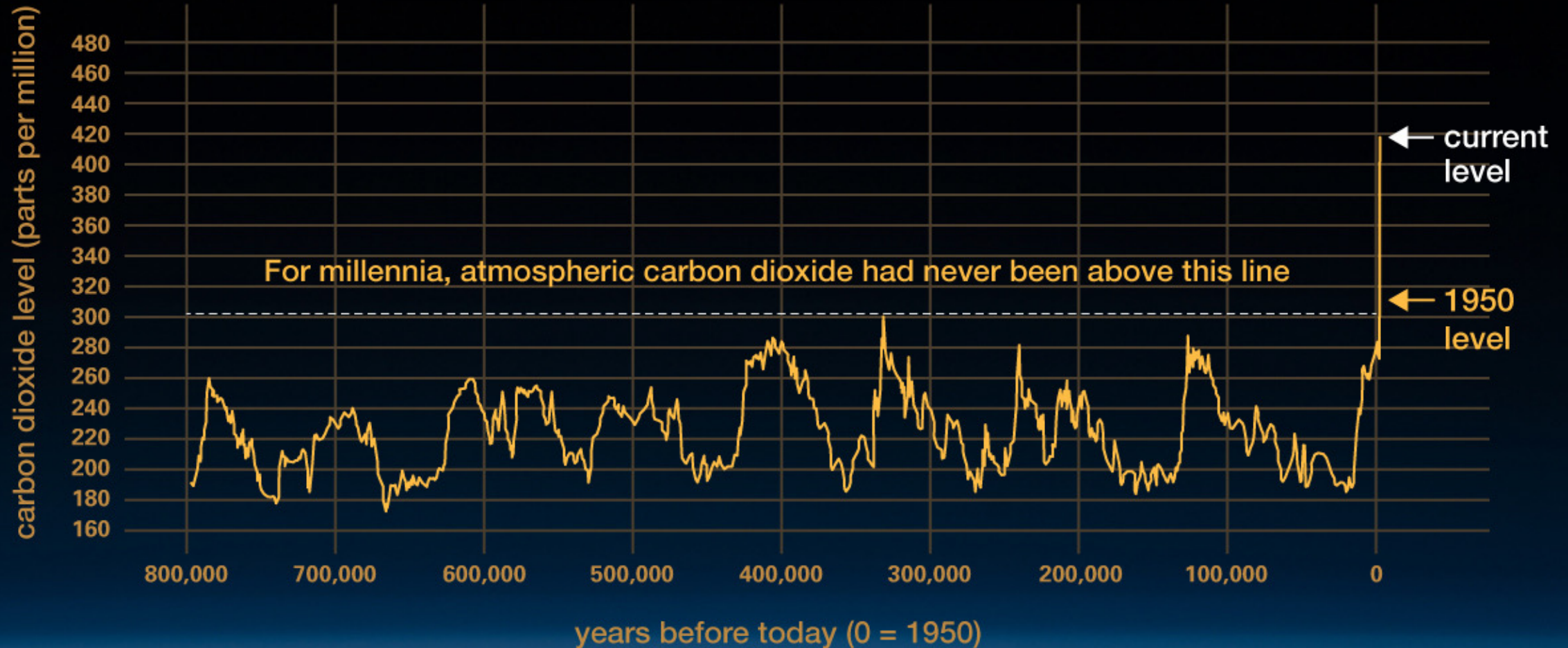
PROGRAMME OF
THE EUROPEAN UNION



Credit: C3S/ECMWF

What really causes global warming?

800,000 Years of Carbon Dioxide

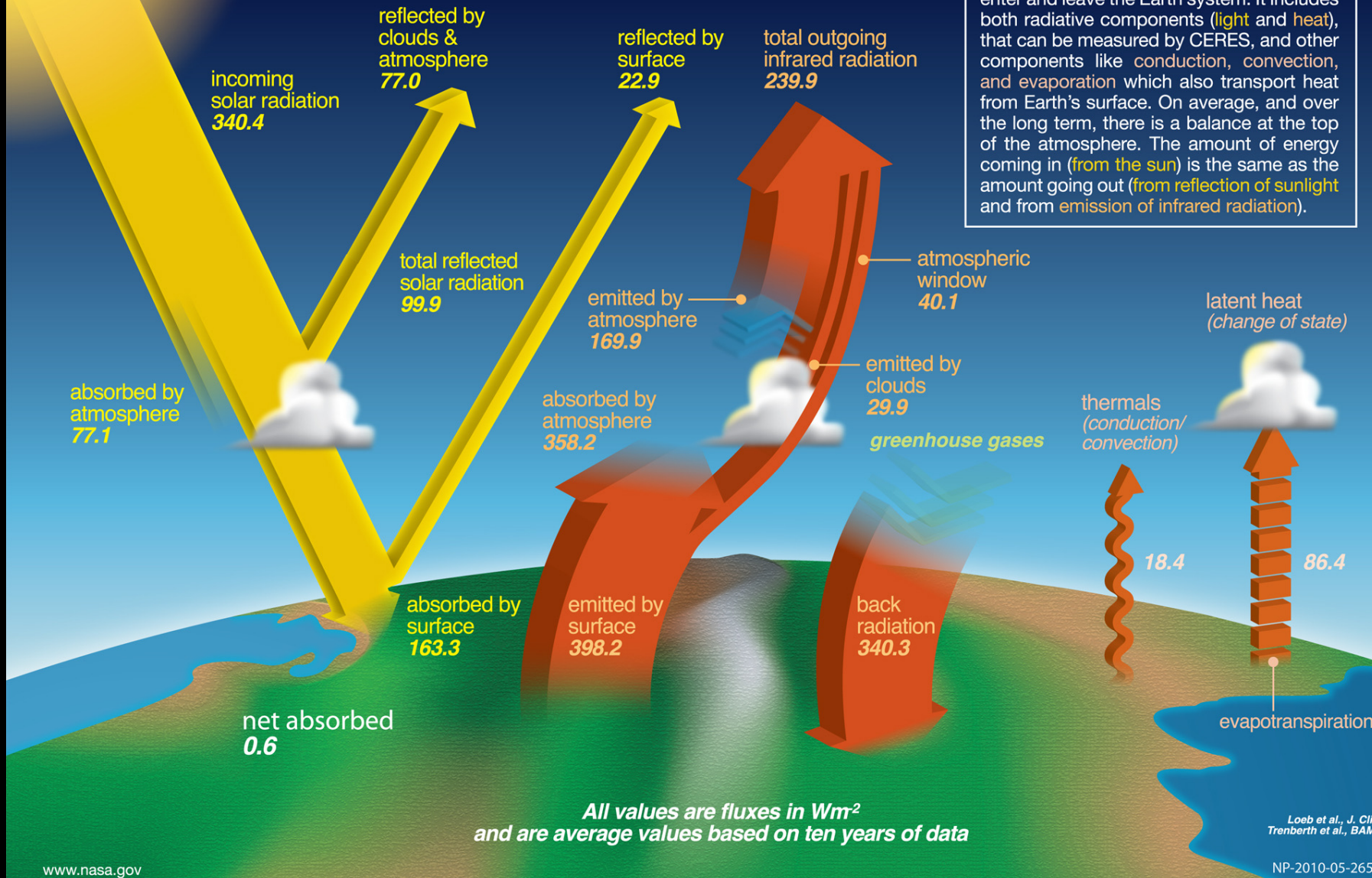




earth's energy *budget*

The Earth's energy budget describes the various kinds and amounts of energy that enter and leave the Earth system. It includes both radiative components (**light** and **heat**), that can be measured by CERES, and other components like conduction, convection, and evaporation which also transport heat from Earth's surface. On average, and over the long term, there is a balance at the top of the atmosphere. The amount of energy coming in (**from the sun**) is the same as the amount going out (**from reflection of sunlight** and from **emission of infrared radiation**).

~0.6W/m² energy imbalance
(receiving more than emitting)



What happens to the heat?

Energy balance model

$$\rho C_p \frac{dT}{dt} = \text{Source} - \text{Sink}$$



Is 0.6W m^{-2} energy imbalance significant

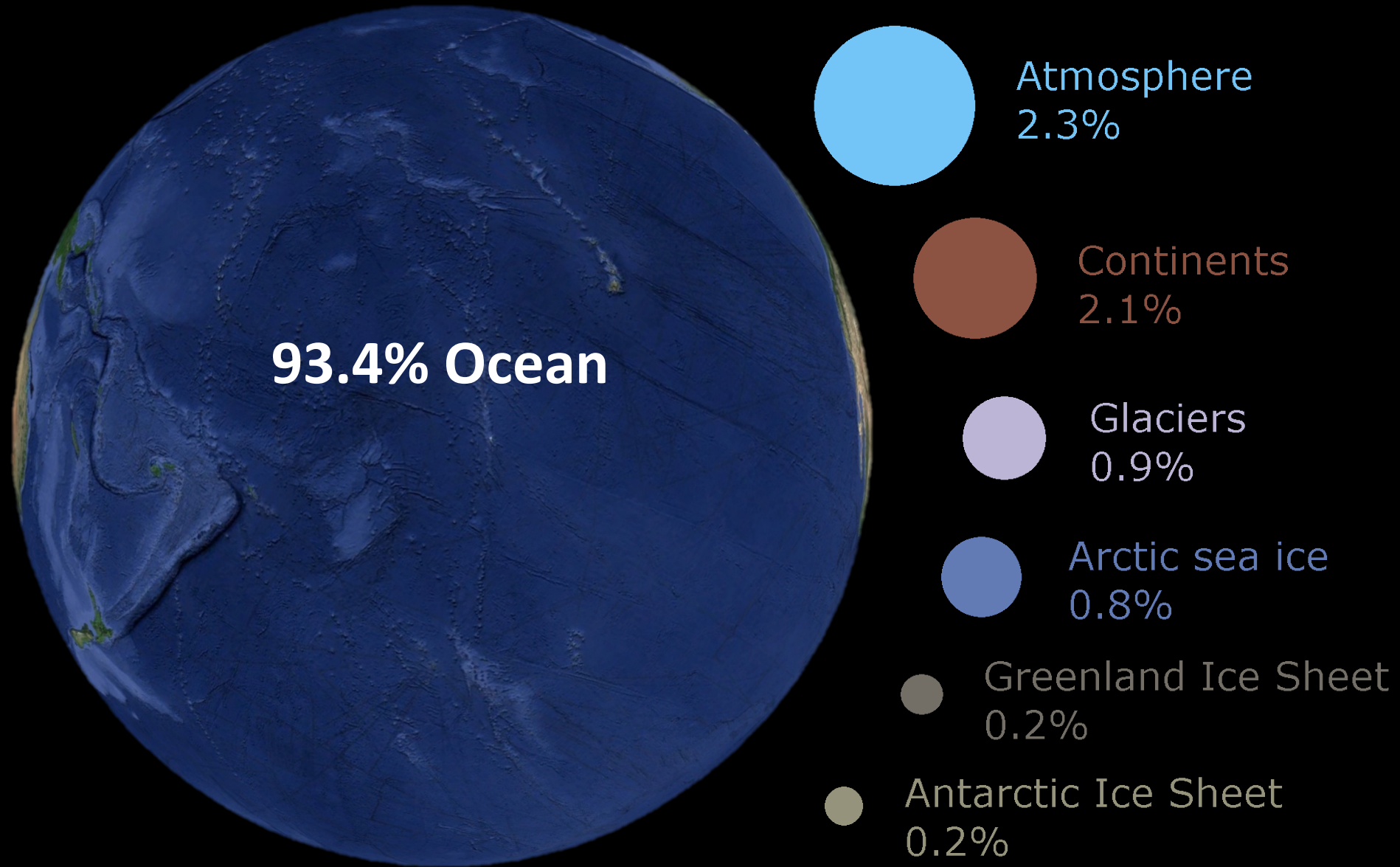
A $\sim 0.6\text{W/m}^2$ energy imbalance might seem small compared to the incoming solar radiation (340.4W/m^2). But is it significant?

It certainly is, given the vast global surface area of Earth: $5.1 \times 10^{12}\text{ m}^2$.

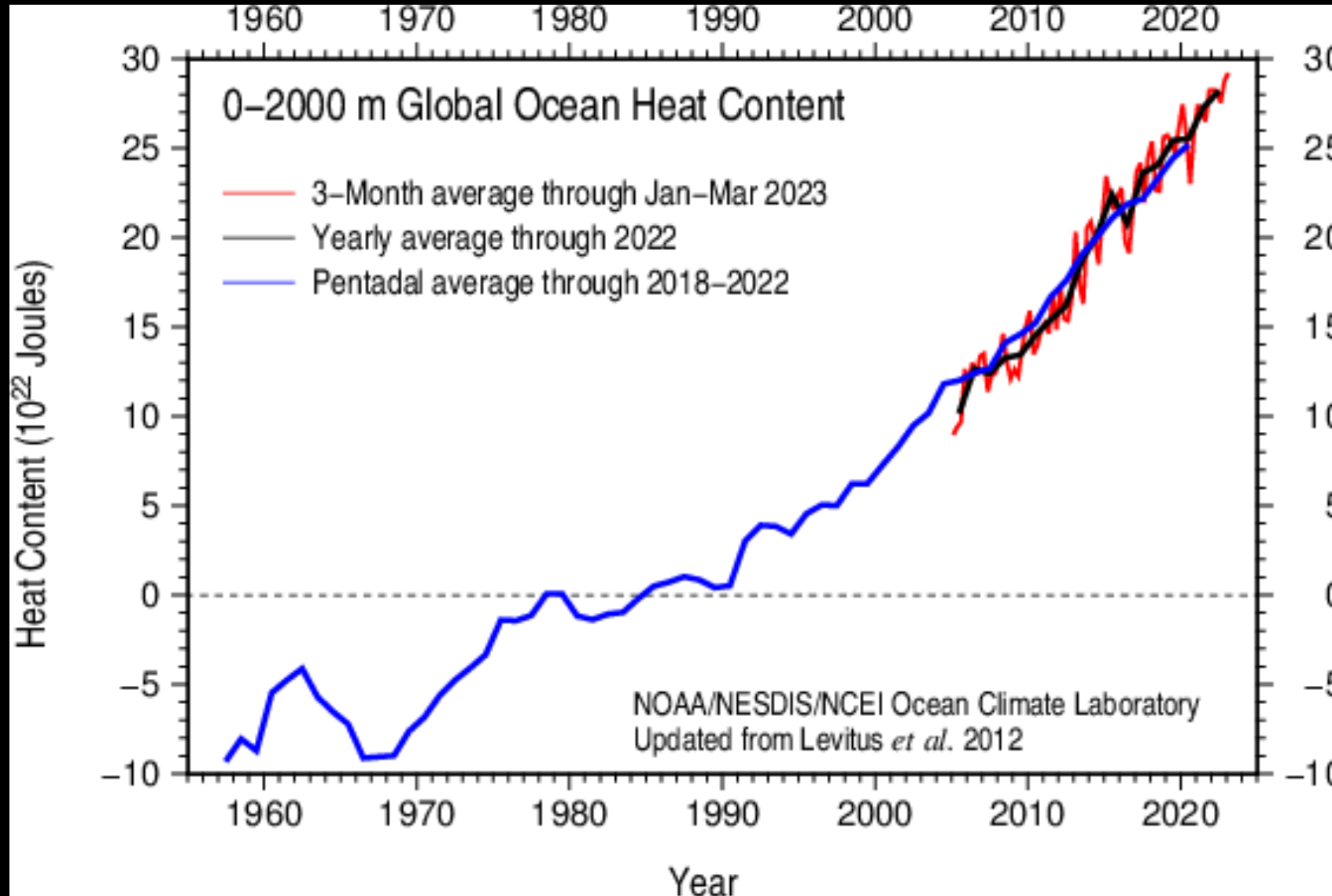
Each year, the global energy imbalance is more than 10 times the total energy consumed globally by humanity.



Where is all the heat going?



Ocean heat has steadily increased

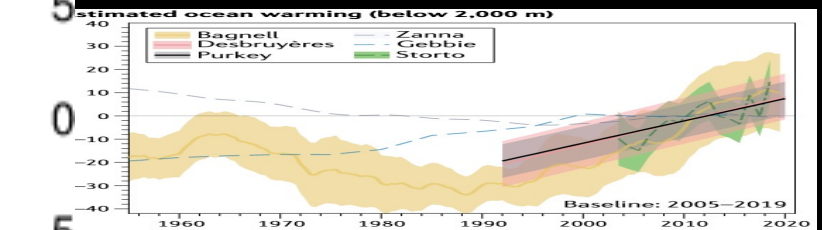


Global ocean heat content has steadily increased. The heat is mainly gained in the upper 2,000 m.

Since 2003, Argo floats have revolutionized the measurements of ocean properties at depth.

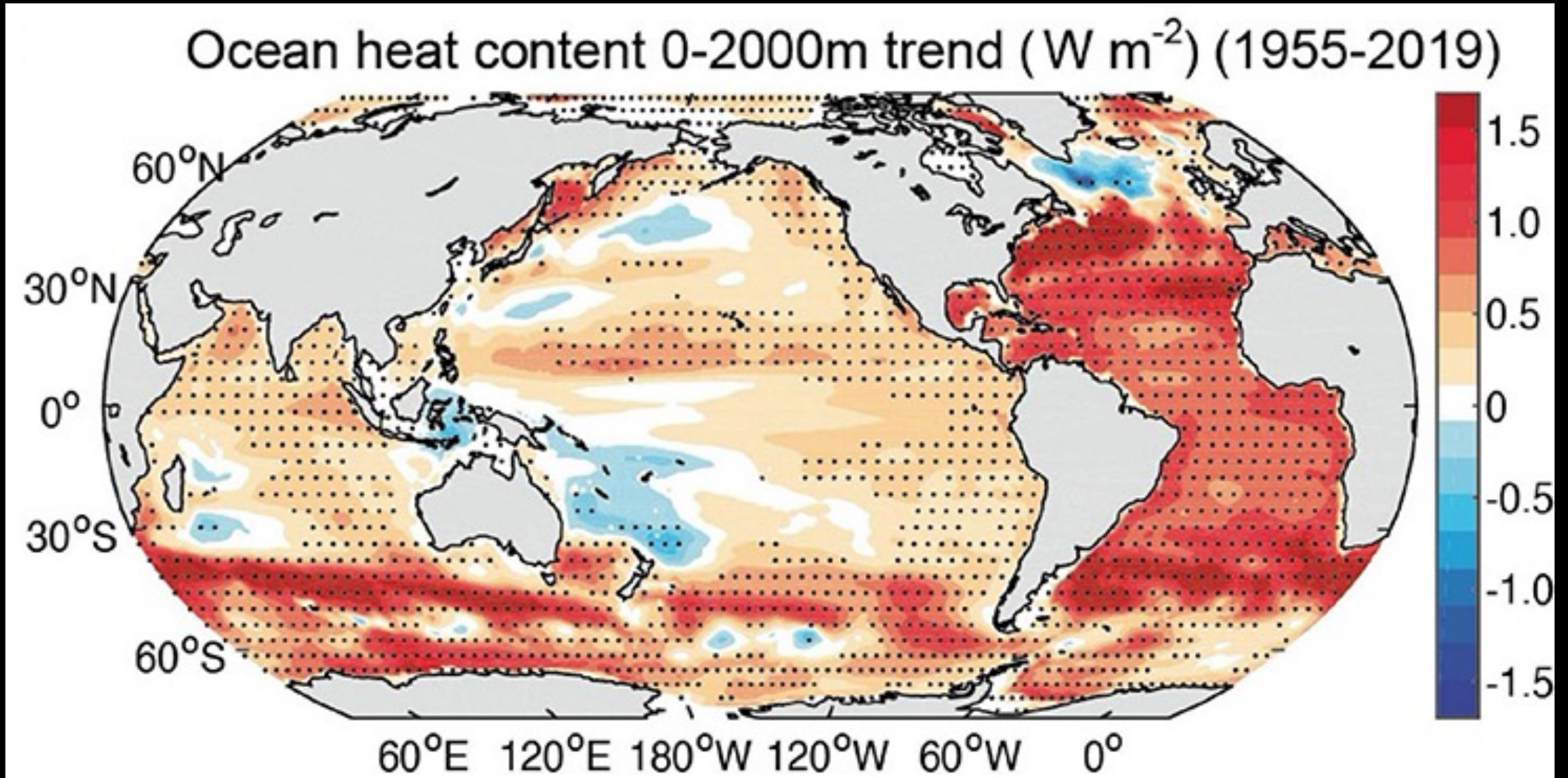


Below 2000 m

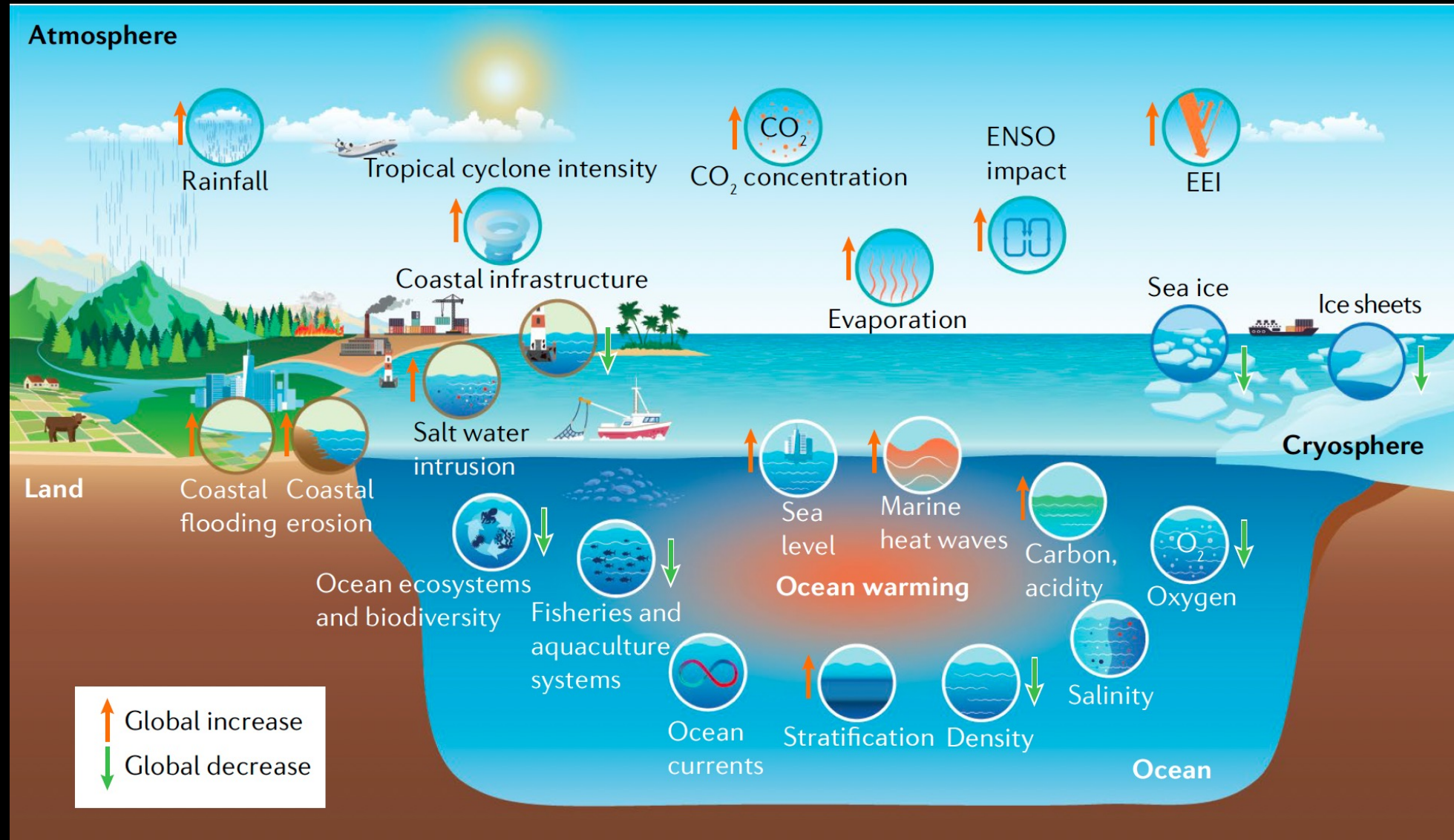


Cheng *et al.* (2022, *Nature*)

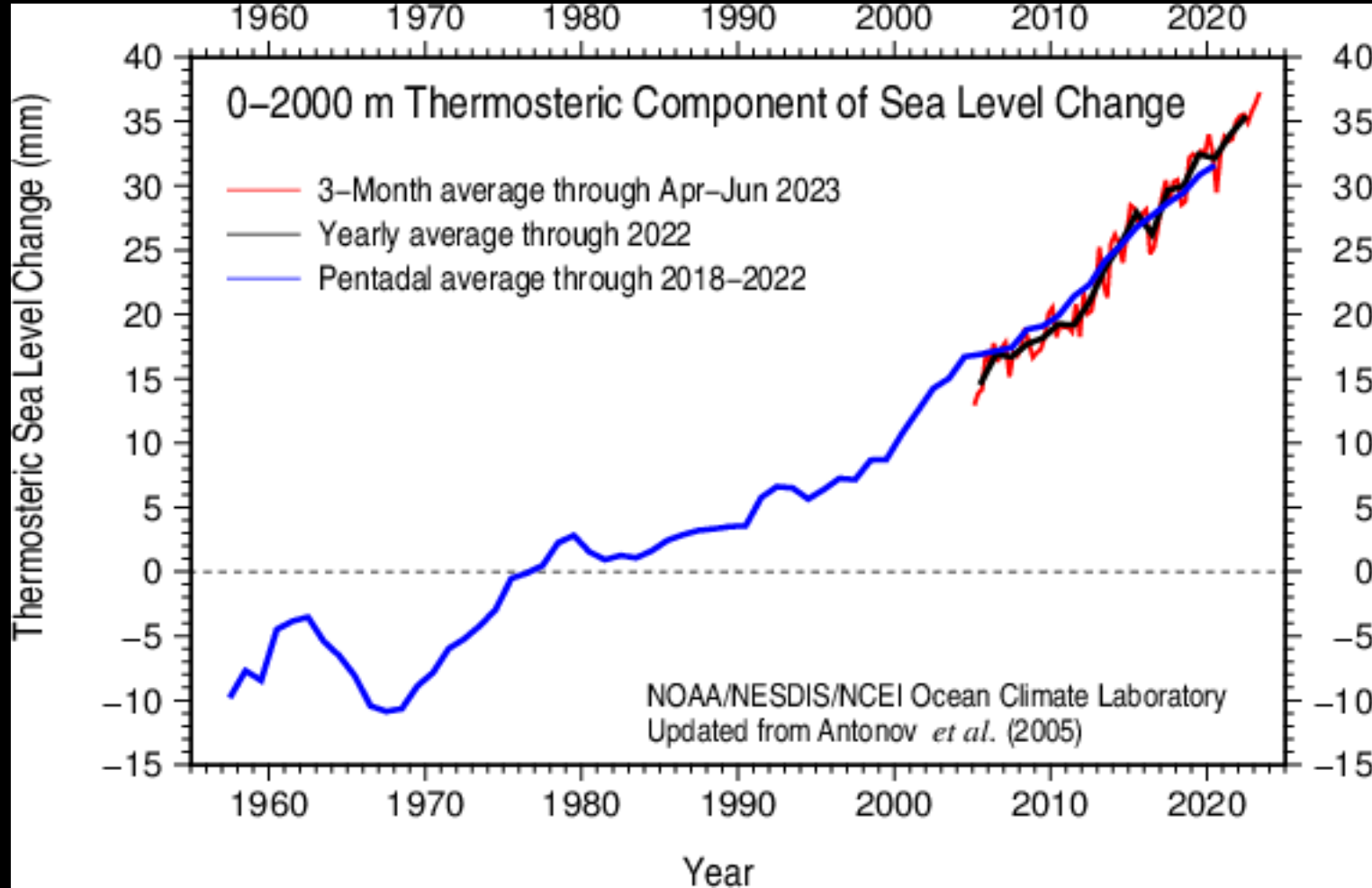
Ocean warming is not uniform



Linkage between ocean warming and changes in Earth system



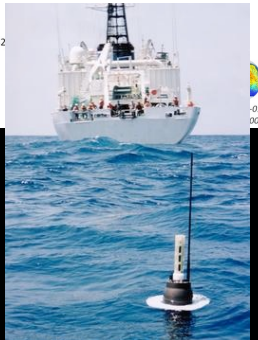
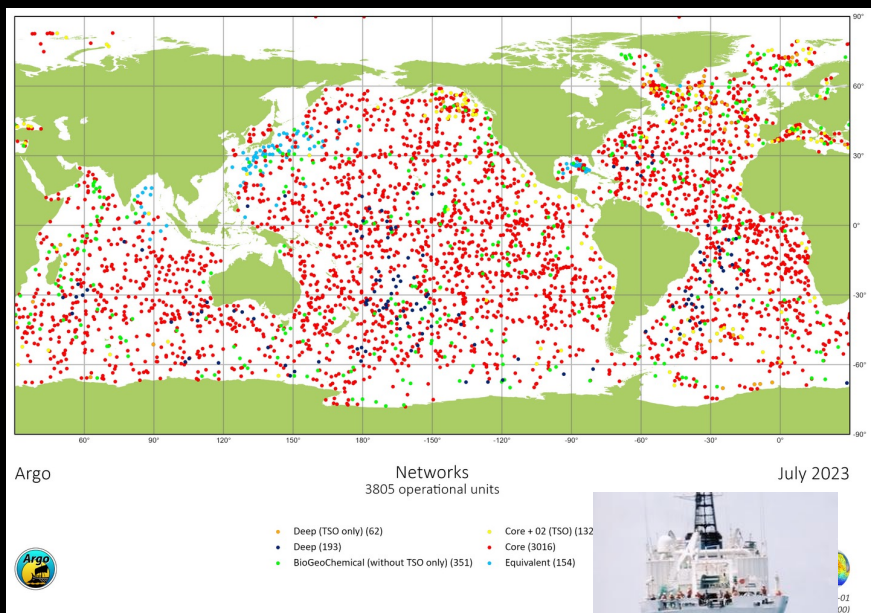
Sea Level Rise, a consequence of global warming



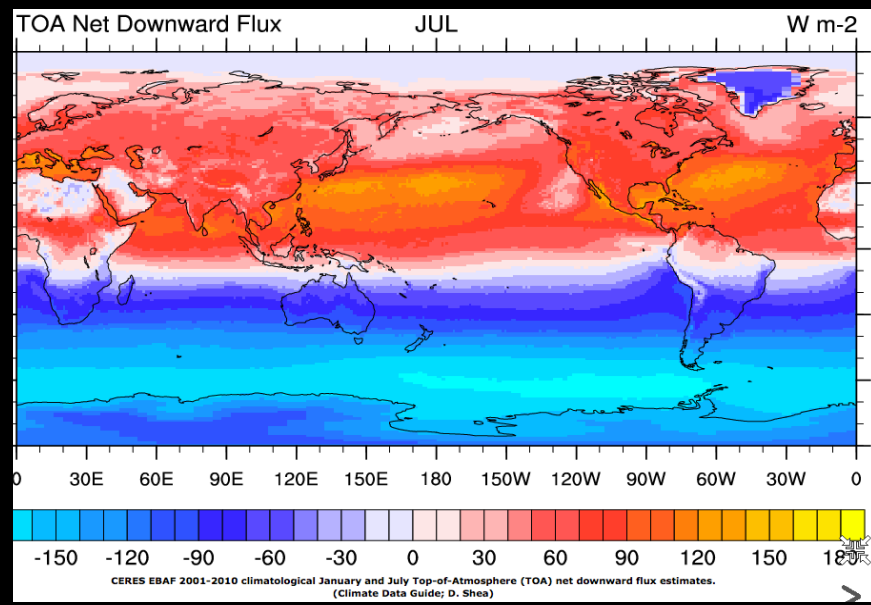
Since 1960: 5.5 inches global sea level rise:

1. The expansion of seawater associated with the warming ocean (thermal expansion) contributed about 1.8 inches (1/3)
2. Melting of ice sheets and mountain glaciers have contributed about 3.7 inches (2/3)

How to quantify where the energy from Earth's net radiative imbalance is accumulating?

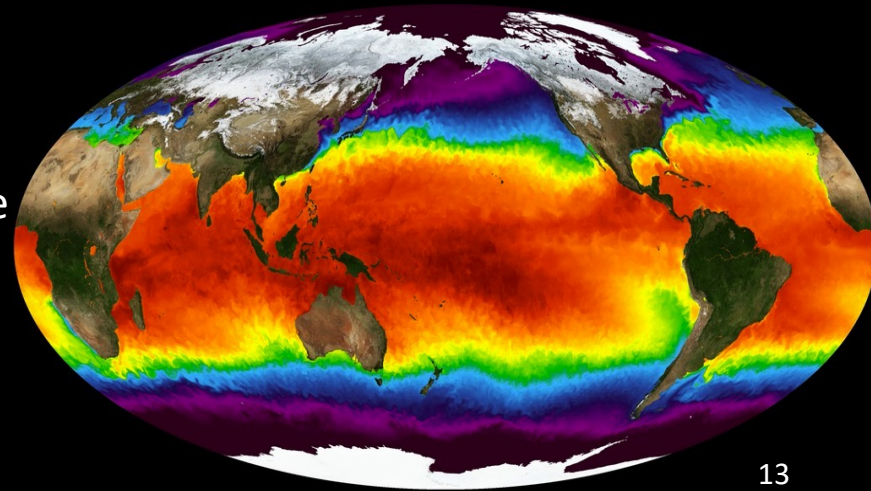


Top of atmosphere radiation
(Energy fluxes)



Argo global array of ocean
profiling floats
(Ocean Interior
Thermal Energy)

Satellite sea surface
temperature
(Earth surface
temperature)



Topic #1 – Where is global warming?

- **Datasets:** ARGO ocean temperature, AMSR-E sea surface temperature, TOA shortwave and longwave
- **Geographic foci:** mid latitudes (+-60-30), low latitudes (+-30-0)
- **Introduction:** In 2015, Earth crossed a major global warming milestone: the global mean surface air temperature (GMSAT) was one degree Celsius warmer than the mid-19th century pre-industrial average. Yet, the road to this 1 degree of atmospheric warming was bumpy: the GMSAT anomaly in 2012 was the same as 2002 and 1998. Because the heat capacity of the atmosphere is much smaller than that of the ocean, much of the year-to-year variability in GMSAT is driven by temperature variations of the ocean surface. If we really want to see where the excess energy of global warming is going, we need to quantify changes in the energy storage in Earth's largest thermal reservoir: the ocean.
- **Questions:**
 1. Global warming is a consequence of an energy imbalance: more shortwave radiation absorbed at the top of the atmosphere (TOA) than re-emitted longwave and reflected shortwave. Calculate the global net radiative flux imbalance at the top of the atmosphere (TOA). How does this compare with published estimates? How has this number changed through time? *Note: when spatially averaging TOA fluxes, make sure you consider the fact that mapped grid cell areas change as a function of latitude.*
 2. If Earth's radiative flux imbalance were entirely absorbed in the troposphere (assume the lower 10 km of atmosphere), what would be the average annual increase in GMSAT? How does your predicted temperature change compare to the actual change through time? *Note: atmospheric density decreases with height.*
 3. Repeat all parts of question (2) but instead consider that the entire radiative flux imbalance warms the upper 10 m, 100 m, 700 m, and 2000 m of the global ocean. Compare the predicted temperature changes against observations by using AMSR-E SST data as a proxy for the upper 10 m ocean temperature, and ARGO data for the upper 100, 700 and 2000 m. How do the actual warming trends of each of these depth categories compare against predictions?
 4. Divide the ocean into 4 basins: Pacific, Indian, Atlantic, and Southern Oceans. Which basins and which depths account for the greatest observed warming?
 5. Compare the year-to-year variations in GMSAT over the past 30 years with variations in the annual mean SST for each of the basins in Question 4. Which basin's SST is the most correlated with GMSAT?
- **Contact Scientist:** Dr. Ou Wang (Ou.Wang@jpl.nasa.gov)